PDF Search

Search, read, annotate and cite your research PDFs. Simple, fast and native to the Mac.

PDF Search is for people with hundreds or thousands of PDFs who want one simple tool: find every paper that mentions something, read it, write notes in its margins, and cite it. Your notes live inside the PDFs and in plain Markdown files, and your bibliography is a plain `library.bib` file. Nothing is locked into this app, and Obsidian or any other notes app is optional.

It runs on macOS 13 (Ventura) or later, on Apple Silicon and Intel Macs.

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1. Installing

1. Open **PDF Search.dmg** and drag **PDF Search** onto the **Applications** folder.
2. Open it from Applications. The first time, macOS warns that it can't verify the app, because it isn't from the App Store or a registered Apple developer. Allow it once:
   - **macOS 15 or later:** click **Done** on the warning (not "Move to Trash"). Open **System Settings → Privacy & Security**, scroll down to *"PDF Search was blocked…"*, click **Open Anyway** and enter your password. Then open the app again and click **Open**.
   - **macOS 13–14:** right-click PDF Search in Applications, choose **Open**, then **Open** again.

> Terminal alternative: `xattr -dr com.apple.quarantine "/Applications/PDF Search.app"`

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2. Adding your PDFs

- **Put your PDFs in `~/Papers`**, a folder called Papers in your home folder. PDF Search indexes it automatically. Keep your own papers in `~/Papers/Library`, and give other big collections (course readings, a conference archive) their own folder inside Papers.
- You can also **drag any other folder or PDF into the window** (or onto the Dock icon), or use **File → Add PDFs or Folders…** (⌘O).
- The first time, PDF Search reads the text of every page. A progress bar shows how far it's got, and you can search while it works. Thousands of papers take a few minutes; after that, searches are instant.

- **Your PDFs stay where they are.** Nothing is copied or moved. PDF Search keeps only an index of their text.
- Folders you add are remembered. Each time the app opens (or when you press ⌘R, *Rescan Folders*), it picks up new and changed PDFs and forgets deleted ones.
- Coming from Zotero? The `tools/import_zotero.py` script in the source code copies your whole library over, highlights included (see its README).

> If a folder syncs with iCloud Drive or Google Drive, make sure its files are actually downloaded to the Mac. "Online-only" files can't be read.

Adding a paper or book

- **Drag a PDF onto the window** (or **+ → Add Paper**, ⌘O). PDF Search finds the paper's DOI (or arXiv ID, ISBN, or its title) and fetches its details. A card shows the type, authors, title, year, journal or book, pages and the citation key it will get. Fix anything, then **Add to Library**: the PDF is moved to `Papers/Library/<key>.pdf` and added to `library.bib`.
- **A book or chapter without a PDF**: **+ → Add Book or Chapter** (⇧⌘O), type the ISBN, press **Look Up**. For a chapter, choose *Book chapter* first, then add the chapter's title, authors and pages.
- **A PDF that has no details yet**: right-click it in the results → **Look Up Citation Details…**.
- Check each card. When there's no DOI, the details come from a title search, which can pick the wrong paper.

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3. Searching

Type in the search box at the top of the sidebar. Results update as you type, best matches first. Each result shows a snippet with your words in bold and how many pages match.

YOU TYPE	IT FINDS
`glitch`	words *starting* with glitch: glitch, glitches, glitching
`glitch feedback`	pages containing *both* words, anywhere on the same page
`"circuit bending"`	that exact phrase
`cafe`	also Café. Accents and capitals don't matter
`art`	art, artifact, artist… but *not* "start" (matches begin at the start of a word)

- File names are searched too (the result says "in name").

- Authors, titles, years and citation keys are searched too.
- **Search in:** the menu above the results limits a search to one of your folders (and folders inside them), or everything. The ••• button next to it can rename a folder (just inside the app), show it in Finder, or remove it from the index.
- With the search box empty, the list shows every PDF in the current folder. Labels: *"no text"* means a scan without a text layer (not searchable); *"can't read"* means damaged or password-protected.
- In the list: click to view, double-click to open in Preview, right-click for *Show in Finder* or *Remove from Index*.

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4. Reading a PDF

- Click a result and the PDF opens at the first match, with every match marked in orange. These marks are temporary and never saved.
- Step through matches with ⌘G / ⇧⌘G or the arrows in the toolbar. The counter shows "3 of 12".
- **Zoom** by pinching on the trackpad. The zoom stays put, even when you open another PDF. ⌘0 or the *Zoom to Fit* button resets it.
- The *Open in Preview* button opens the file in your usual PDF app.

Find inside the open PDF (⌘F)

Already found the right paper and want to look for something else in it? Press ⌘F. A find bar appears above the PDF; whatever you type is marked in that PDF (anywhere in a word, like Preview). Return or ⌘G goes to the next match, ⇧⌘G to the previous one. Esc or **Done** closes the bar and brings back your main search's marks.

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5. Highlights and notes

This is how you read with a pencil, and there's only one thing to learn: **select a passage and press H**. You can also click the big **Highlight selected text** button that appears in the notes column whenever text is selected. The passage is highlighted, and the cursor jumps into a note box beside it, so you can just start typing. The note saves itself as you type, and Return starts a new line. If you only wanted the highlight, click anywhere else.

- The **Highlights & Notes** column on the right lists every highlight in page order. Each card shows its page, the passage, and its note box.

- **Page 12** (the blue link on a card) jumps to the passage. Clicking a highlight in the PDF selects its card.
- **Remove** (trash icon) deletes a highlight and its note.
- **About this paper**, the box at the top of the column, is for notes on the whole paper: a summary, the main argument, how you might use it. It saves itself, is stored in the PDF (as a note on page 1), and appears as a Summary section in the paper's Markdown note.
- The coloured dots at the top of the column choose the colour of new highlights. The **Notes** button in the toolbar shows or hides the column.
- Highlights and notes are saved **inside the PDF** as standard PDF comments, so Preview, Acrobat and other apps show them too. Notes are also searchable, so you can find your own thoughts across every paper.

Paper notes (Markdown)

Every PDF you highlight also gets a Markdown note named after its citation key, e.g. `@cascone2000aesthetics.md`. It lists each passage you marked as a quote, with its page, a link that opens PDF Search at that page, a ready-made citation `[@cascone2000aesthetics, p. 3]`, and your note underneath. Write your own thoughts *below* the marginalia section; PDF Search only ever rewrites the marginalia section itself. Notes go to `~/Papers/Notes`, or `Literature` in your vault if you have a folder called *Obsidian Vault*. Any notes app or text editor can open them. **File → Write Notes for All Annotated PDFs** creates notes for PDFs you highlighted before.

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6. Your library and citations

- Your bibliography is one plain text file, `library.bib` (in `~/Papers`, or in your Obsidian vault). Each paper or book has a **citation key**: last name + year + first meaningful word of the title, e.g. `cascone2000aesthetics`. A PDF in `Library` named `<key>.pdf` is matched to its entry automatically.
- Results then show the real title, "Cascone (2000)" and the key. Books you don't have as PDFs appear too, as a card with their details.
- **Cite**: `⇧⌘C` copies `[@key]` (Markdown/Pandoc), `⌥⇧⌘C` copies `\cite{key}` (LaTeX/Overleaf). The same choices are in the toolbar's **Cite** button and the right-click menu.
- Writing in Markdown: cite with `[@key]`, then export with Pandoc (`pandoc paper.md --citeproc --bibliography library.bib -o paper.pdf`). Writing in LaTeX/Overleaf: upload `library.bib` and use `\cite{key}`.

7. Collections

A collection is a reading list for something you're writing: a paper, a chapter, a syllabus. It's saved as a plain Markdown note (in `Collections` in your Obsidian vault, or `~/Papers/Collections`), one line per item: `- [[@cascone2000aesthetics]] Cascone (2000) The Aesthetics of Failure`.

- **Add** the open paper with **Add to Collection** in the toolbar (also in the right-click menu of results, and on a book's card). **New Collection…** in that menu creates one.
- **Browse**: choose the collection in **Search in**. You see just its papers and books, in the note's order, and searching stays inside it. Right-click → **Remove from "…"** takes an item out.
- **Make Bibliography** (••• next to Search in) writes `<collection>.bib` with just those entries, ready for Overleaf or Pandoc.
- The note is yours to edit: reorder lines, add headings, write comments. PDF Search reads every `@key` in it.

8. Keyboard shortcuts

KEYS	ACTION
⌘O / ⇧⌘O	Add a paper (PDF) / add a book or chapter (ISBN)
⌘R	Rescan folders for new or changed PDFs
⌘F	Find inside the open PDF
⌘G / ⇧⌘G	Next / previous match
H	Highlight the selected text and write a note beside it (also ^⌘H)
⇧⌘C / ⌥⇧⌘C	Copy citation `[@key]` / `\cite{key}`
⌘,	Settings: papers folder, library.bib, notes folder
⌘0	Zoom to fit

9. Good to know

- **Scanned PDFs** (pictures of pages) have no text to search until they're run through OCR, e.g. with Acrobat or a free OCR tool. They're labelled "no text".
- All search words must be on the **same page**.

- If you move or rename a folder you've added, remove it in the app (••• → Remove Folder from Index) and drag it in again.
- **Settings (⌘,)** let you choose the papers folder, where `library.bib` lives, and where paper notes and collections go (or turn paper notes off).
- **Where the index lives:** `~/Library/Application Support/PDF Search/`. **File → Clear Index…** empties it; your PDFs are untouched.
- **To uninstall:** delete PDF Search from Applications, and optionally the folder above.

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10. Extras in the download

- **Source Code:** the entire app is one Swift file. To build it yourself (no warnings on first launch), install Apple's command line tools (`xcode-select --install`) and double-click *Build PDF Search.command*. *Developer Notes.md* explains how it works and how to extend it.
